Study Notes: Matrices and Systems of Equations

1 Systems of Linear Equations

A **system of linear equations** is a set of equations involving the same set of variables. For example:

$$\begin{cases} 2x + 3y = 5 \\ 4x - y = 11 \end{cases}$$

This can be written in matrix form as:

$$A\mathbf{x} = \mathbf{b} \quad \text{where} \quad A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

2 Row Echelon Form (REF)

A matrix is in **row echelon form** if:

- All nonzero rows are above any all-zero rows.
- The leading entry of each nonzero row is to the right of the one in the row above it.
- Entries below a leading entry are all zero.

The **reduced row echelon form (RREF)** also requires:

- Leading entries are 1.
- Each leading 1 is the only nonzero entry in its column.

Example: Row operations to reach RREF

Consider the matrix
$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ -1 & 0 & 2 \end{bmatrix}.$$

We will use elementary row operations to reduce A to reduced row echelon form (RREF).

Step 1: Use the leading 1 in row 1 to clear entries below it

We want zeros below the leading 1 in the first column.

Replace row 2 by row 2 minus 2 times row 1:
$$R_2 \leftarrow R_2 - 2R_1.$$

Replace row 3 by row 3 plus row 1:
$$R_3 \leftarrow R_3 + R_1.$$

This gives
$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}.$$

Step 2: Make the pivot in row 2 equal to 1

Currently row 2 has a leading entry of -1 in the second column. Multiply row 2 by -1:
$$R_2 \leftarrow -1 \cdot R_2.$$

Now we have
$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 2 & 1 \end{bmatrix}.$$

Step 3: Use the pivot in row 2 to clear the entry below it

We want a zero below the pivot (the 1 in row 2, column 2). Replace row 3 by row 3 minus 2 times row 2:

$$R_3 \leftarrow R_3 - 2R_2.$$

This gives

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 7 \end{bmatrix}.$$

Step 4: Make the pivot in row 3 equal to 1

Row 3 has a leading entry of 7 in the third column. Multiply row 3 by $\frac{1}{7}$:

$$R_3 \leftarrow \tfrac{1}{7}R_3.$$

Now the matrix is

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

At this point the matrix is in row echelon form. Next we clear the entries above each pivot to reach reduced row echelon form.

Step 5: Clear entries above the pivot in column 3

Use the pivot in row 3, column 3 (the 1) to make the entries above it zero.

Row 1 operation:

$$R_1 \leftarrow R_1 + R_3 \quad \text{(because row 1 has } -1 \text{ in column 3).}$$

Row 2 operation:

$$R_2 \leftarrow R_2 + 3R_3 \quad \text{(because row 2 has } -3 \text{ in column 3).}$$

The matrix becomes

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 6: Clear entries above the pivot in column 2

Use the pivot in row 2, column 2 to make the entry above it zero. Replace row 1 by row 1 minus 2 times row 2:

$$R_1 \leftarrow R_1 - 2R_2.$$

The final matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which is the reduced row echelon form (RREF) of A.

3 Matrix Arithmetic

Matrix operations include:

- **Addition/Subtraction:** Only if dimensions match.
- **Scalar multiplication:** Multiply each element by a constant.
- **Matrix multiplication:** If A is $m \times n$, B must be $n \times p$. The result is $m \times p$.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

4 Matrix Algebra

Important properties:

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- Identity matrix I: $AI = IA = A$
- Inverse matrix A^{-1}: $AA^{-1} = A^{-1}A = I$

5 Elementary Matrices

An **elementary matrix** results from applying one row operation to the identity matrix.

Types of elementary row operations:

- Swap two rows.
- Multiply a row by a nonzero scalar.
- Add a multiple of one row to another.

Multiplying a matrix A on the left by an elementary matrix E applies the corresponding row operation to A.

6 Partitioned Matrices

A **partitioned matrix** is divided into smaller submatrices (blocks).

Example:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

Block matrices are useful in simplifying large matrix operations and in applications like parallel computing and system design.

Matrix Multiplication

Matrix multiplication is a way of combining two matrices to produce a third matrix. It is defined when the number of *columns* of the first matrix equals the number of *rows* of the second matrix.

Definition

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix},$$

where A is an $m \times n$ matrix and B is an $n \times p$ matrix.

Then the product $C = AB$ is an $m \times p$ matrix whose (i, j)-entry is

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words: the entry c_{ij} in row i, column j of the product is obtained by taking the dot product of row i of A with column j of B.

Important facts

- The product AB is defined only if the number of columns of A equals the number of rows of B.
- The size of AB is $m \times p$ (rows of A by columns of B).
- Matrix multiplication is *not* commutative in general: usually $AB \neq BA$.
- Matrix multiplication is associative: $(AB)C = A(BC)$ whenever the products are defined.
- Matrix multiplication is distributive over addition: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.

Worked example: Multiplying two 2×2 matrices

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}.$$

We want to compute AB.

Since both A and B are 2×2, the product AB will also be a 2×2 matrix:

$$AB = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Entry c_{11}: row 1 of A times column 1 of B:

$$c_{11} = 1 \cdot 2 + 2 \cdot (-1) = 2 - 2 = 0.$$

Entry c_{12}: row 1 of A times column 2 of B:

$$c_{12} = 1 \cdot 0 + 2 \cdot 5 = 0 + 10 = 10.$$

Entry c_{21}: row 2 of A times column 1 of B:

$$c_{21} = 3 \cdot 2 + 4 \cdot (-1) = 6 - 4 = 2.$$

Entry c_{22}: row 2 of A times column 2 of B:

$$c_{22} = 3 \cdot 0 + 4 \cdot 5 = 0 + 20 = 20.$$

Therefore,

$$AB = \begin{bmatrix} 0 & 10 \\ 2 & 20 \end{bmatrix}.$$

Worked example: Multiplying a 2×3 matrix by a 3×2 matrix

Now consider

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \quad (2 \times 3 \text{ matrix})$$

and

$$B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 4 \end{bmatrix} \quad (3 \times 2 \text{ matrix}).$$

The product AB is defined because the number of columns of A (which is 3) equals the number of rows of B (which is 3). The result will be a 2×2 matrix.

Write

$$AB = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}.$$

Entry d_{11}: row 1 of A times column 1 of B:

$$d_{11} = 1 \cdot 2 + (-1) \cdot (-1) + 2 \cdot 3 = 2 + 1 + 6 = 9.$$

Entry d_{12}: row 1 of A times column 2 of B:

$$d_{12} = 1 \cdot 1 + (-1) \cdot 0 + 2 \cdot 4 = 1 + 0 + 8 = 9.$$

Entry d_{21}: row 2 of A times column 1 of B:

$$d_{21} = 0 \cdot 2 + 3 \cdot (-1) + 1 \cdot 3 = 0 - 3 + 3 = 0.$$

Entry d_{22}: row 2 of A times column 2 of B:

$$d_{22} = 0 \cdot 1 + 3 \cdot 0 + 1 \cdot 4 = 0 + 0 + 4 = 4.$$

So

$$AB = \begin{bmatrix} 9 & 9 \\ 0 & 4 \end{bmatrix}.$$